

# ABDOUL TOURE

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## SUMMARY

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Results-driven Senior Data Scientist with 10+ years of experience designing and deploying machine learning models, LLM-powered pipelines, and scalable data solutions. Proven record leading full life-cycle data science projects, managing cross-functional collaboration, and delivering predictive models that boost ROI, reduce costs, and streamline operations. Highly proficient translating complex data into actionable insights, applying prompt engineering and Vertex AI generative models, developing automated ML pipelines, and mentoring teams to foster data-driven decision-making. Currently advancing expertise through PhD studies in Data Science, deepening research capabilities and strengthening model development for real-world applications.

## TECHNICAL SKILLS

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**Python:** PySpark, Requests/REST APIs, scikit-learn, NLTK, SciPy, JSON, PyTest, Hadoop/HDFS, ODBC, Apache, Hive, Snowflake, Oracle, PostgreSQL, Linux, R, SAS, Tableau, Regex, Power BI, BigQuery, feature engineering for large datasets, ETL and data pipeline design

**Cloud and AI Platforms:** Google Cloud Vertex AI, LLM pipeline development, prompt engineering, Microsoft Azure, Microsoft Fabric (OneLake, Synapse Data Science, MLflow)

### AI/ML Frameworks

**Classification:** Binary, multi-class, multi-label, propensity modeling, intent prediction

**Regression:** OLS, logistic, multivariate, penalized Lasso/Ridge, Poisson, negative binomial, tree-based models, XGBoost, random forest

**Clustering:** K-Means, hierarchical clustering, segmentation models, t-SNE

**Forecasting and Validation:** ARIMA, Prophet, ML-based time series, model validation and diagnostics, SHAP feature importance

**NLP / LLMs:** Topic modeling, embeddings, transcript mining, sentiment analysis, RAG workflows, prompt engineering, digital message optimization

**Causal and Statistical Methods:** Difference-in-differences, interrupted time series, propensity score matching, causal inference for observational data, design of experiments, power analysis

**Operations and Call Center Modeling:** Queueing theory (M/M/1, M/M/k), call volume forecasting, staffing optimization, performance modeling focused on AHT and FCR, simulation for shifting call traffic to digital tools

**Behavioral Science:** Nudges, defaults, habit-formation modeling, churn and retention analytics, digital adoption uplift modeling

## EXPERIENCE

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**CVS Health - Analytics and Behavioral Change Org - Caremark Digital Call Reduction and Digital Adoption** Remote  
**Senior Data Scientist** 2025-Present

- Lead data wrangling, ETL, and analysis of 60M+ call center and digital adoption records to identify drivers of calls and improve self-service.
- Build SQL/BigQuery dashboards and predictive models to reduce call volume, cost, and average handle time (AHT).
- Apply statistical modeling (regression, quasi-experimental methods, survival analysis) to measure impact of digital tools and interventions.
- Use queueing theory and forecasting to optimize staffing and predict call demand.
- Analyze call transcripts with NLP/LLM methods (topic modeling, embeddings, sentiment analysis) to enhance customer messaging and digital engagement.
- Develop and deploy LLM-powered pipelines on Google Cloud Vertex AI to automate large-scale analysis of member call transcripts and digital interactions, surfacing behavioral patterns that inform both marketing and operational strategy.
- Design and iterate on prompt engineering frameworks for Vertex AI generative models, optimizing output quality, consistency, and safety for healthcare member communication and digital engagement content.
- Apply K-Means and hierarchical cluster analysis to segment 60M+ member records by call behavior, digital adoption propensity, and self-service engagement patterns, enabling precision targeting of demand generation and retention interventions.

- Incorporate behavioral science principles (nudges, habit formation) into models that encourage sustained digital adoption.

## AMSIVE

Remote

### Senior Data Scientist

2022-2025

- Leveraged Microsoft Fabric to build end-to-end data science solutions, designing and deploying ML models (XGBoost, logistic regression) that boosted campaign ROI by 25% and cut acquisition costs by 18%.
- Engineered scalable data pipelines within Microsoft Fabric's OneLake using PySpark and SQL to process over 50M records, reducing data preparation time by 40%.
- Led the development of 15+ models from conception to deployment on Fabric's unified platform, improving project delivery speed and cross-functional collaboration.
- Increased model performance by 10% (ROC-AUC) through advanced feature engineering and Fabric's integrated Synapse Data Science and MLflow capabilities for experiment tracking.

## WELLS FARGO

San Antonio, Texas

### Lead Analytics Consultant

2020-2022

- Conducted data mining across multiple sources, enhancing data accuracy by 25% and reducing processing time by 30%.
- Identified fraud patterns through advanced statistical analysis, improving actionable insight-gathering by 5%.
- Applied statistical tools to improve data interpretation and reduce analytical errors by 10%.
- Provided strategic recommendations, mentored junior analysts, and increased project success rates by 20%.

## WELLS FARGO

San Antonio, Texas

### Analytics Consultant

2018-2020

- Developed dashboards and KPIs in Tableau and Power BI, improving reporting accuracy by 15%.
- Standardized analytics frameworks across regulatory teams to streamline compliance reporting.
- Enhanced reporting environments, leading to a 30% performance trend improvement.
- Ensured risk policy adherence, reducing data discrepancies by 25%.

## HARLAND CLARKE

San Antonio, Texas

### Analytics Manager

2017-2018

- Managed 20+ client analytics projects, delivering insights for digital marketing and e-commerce optimization.
- Built marketing attribution models that increased ROI by 15% and engagement by 10%.
- Developed audience segmentation frameworks, enhancing campaign targeting by 20%.
- Partnered with BI and marketing teams on A/B testing and model improvements.

## HARLAND CLARKE

San Antonio, Texas

### Technical Risk Analyst II

2016-2017

- Conducted full life-cycle modeling for marketing strategies, resulting in a 15% lift in conversions.
- Led cross-sell and upsell initiatives that generated a 10% increase in client revenue.
- Delivered projects with custom data integrations, reducing delivery time by 20%.

## PEARSON

San Antonio, Texas

### Research Analyst

2015-2016

- Provided psychometric support and statistical programming for national assessments.
- Automated SAS-based data cleaning and validation processes, cutting processing time by 25%.
- Established quality control procedures across educational data pipelines.

## ADDITIONAL EXPERIENCE

UNITED STATES AIR FORCE, San Antonio, Texas. Technical Sergeant, Active Duty / Texas ANG / Texas ARNG, 2009-2016. Led mission planning and operations across multidisciplinary teams in high-pressure environments. Delivered technical training and mentorship on advanced equipment and systems. Demonstrated resilience and leadership in both domestic and deployed operations.

## EDUCATION

### NATIONAL UNIVERSITY, San Diego, California

Ph.D. Student in Data Science, 2025-Present

**UNIVERSITY OF TEXAS, San Antonio, Texas**

M.S. in Statistics, GPA: 3.8, 2017

B.S. in Actuarial Science, GPA: 3.9, 2015

**CERTIFICATIONS**

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Microsoft Certified: Azure Data Scientist Associate

Designing and Implementing Data Science Solutions on Azure